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1η Σειρά Ανάλυτικών Ασκήσεων
Νευρωνικά Δίκτυα και Βαθιά Μάθηση

Άσκηση 1 (Multi Layer Perceptron)

a) $x_1 = -1$, $x_2 = 1$

$\Delta w_{03} = d_3$, $\Delta w_{04} = d_4$

$\Delta w_{kj} = -\epsilon \frac{\partial E}{\partial a_k} \frac{\partial a_k}{\partial net_k} \frac{\partial net_k}{\partial w_{kj}}$, όπου $\beta_k = w_{0k} + \sum_i x_i \cdot w_{ik}$

και $a_k = g(\beta_k)$

$\Delta w_{kj} = -\epsilon \frac{\partial E}{\partial w_{kj}} = -\epsilon \frac{\partial E}{\partial \beta_k} \frac{\partial \beta_k}{\partial w_{kj}}$, επομένως

$\Delta w_{0k} = -\epsilon \frac{\partial E}{\partial \beta_k} \frac{\partial \beta_k}{\partial w_{0k}} = -\epsilon \frac{\partial E}{\partial \beta_k}$

$\Delta w_{ij} = -\epsilon \frac{\partial E}{\partial \beta_j} \frac{\partial \beta_j}{\partial w_{ij}} = -\epsilon \cdot x_i \frac{\partial E}{\partial \beta_j}$

$\Delta w_{03} = -\epsilon \frac{\partial E}{\partial \beta_3} = d_3$, $\Delta w_{04} = -\epsilon \frac{\partial E}{\partial \beta_4} = d_4$

$\Delta w_{13} = -\epsilon \frac{\partial E}{\partial \beta_3} \cdot x_1 = -d_3$

$\Delta w_{23} = -\epsilon \frac{\partial E}{\partial \beta_3} x_2 = d_3$

$\Delta w_{14} = -\epsilon \frac{\partial E}{\partial \beta_4} x_1 = -d_4$

$\Delta w_{24} = -\epsilon \frac{\partial E}{\partial \beta_4} x_2 = d_4$

b) $x_1 = -1, x_2 = 1$

$t_5 = 1$

$w_{03} = 1, w_{13} = -1, w_{23} = -1$

$w_{04} = -2, w_{14} = -1, w_{24} = -1$

$w_{05} = 2, w_{35} = -1, w_{45} = 1$

$\epsilon = 1.0$

$g(x) = 1/(1+e^{-x})$

$b_3 = w_{03} + x_1 w_{13} + x_2 w_{23} = 1 + 1 - 1 = 1$

$a_3 = g(b_3) = 0,731$

$b_4 = w_{04} + x_1 w_{14} + x_2 w_{24} = -2 + 1 - 1 = -2$

$a_4 = g(b_4) = 0,119$

$b_5 = w_{05} + a_3 w_{35} + a_4 w_{45} = 2 - 0,731 + 0,119 = 1,388$

$\boxed{a_5 = g(b_5) = 0,8}$

$\boxed{E = \frac{1}{2}(t_5 - a_5)^2 = 0,02}$

$\Delta w_{05} = -\epsilon \frac{\partial E}{\partial b_5} \overset{\epsilon=1.0}{=} -\frac{\partial E}{\partial a_5}\frac{\partial a_5}{\partial b_5} = -(t_5 - a_5)\cdot g'(b_5) =$

g sigmoid

$= -0,2 \cdot g(b_5)(1 - g(b_5)) = -0,2 \cdot 0,8 \cdot 0,2 = -0,032$

$\Delta w_{35} = -\epsilon x_3 \frac{\partial E}{\partial b_5} = -0,731 \cdot 0,032 = -0,023$

$\Delta w_{45} = -\epsilon a_4 \frac{\partial E}{\partial b_5} = -0,119 \cdot 0,032 = -0,004$

$\Delta w_{03} = -\epsilon \frac{\partial E}{\partial b_3} = -\frac{\partial E}{\partial a_3}\frac{\partial a_3}{\partial b_3} = -\frac{\partial E}{\partial b_5}\frac{\partial b_5}{\partial a_3}\frac{\partial a_3}{\partial b_3} = -0,032 \cdot$

$w_{35} \cdot g'(b_3) = -0,032 \cdot g(b_3)(1 - g(b_3)) = 0,006$

$\Delta w_{04} = -\epsilon \frac{\partial E}{\partial b_4} = -\frac{\partial E}{\partial a_4}\frac{\partial a_4}{\partial b_4} = -\frac{\partial E}{\partial b_5}\frac{\partial b_5}{\partial a_4}\frac{\partial a_4}{\partial b_4} = -0,032 \cdot$

$w_{45} \cdot g'(b_4) = -0,032\, g(b_4)(1 - g(b_4)) = 0,003$

$\Delta w_{13} = -0,006$, $\Delta w_{23} = 0,006$, $\Delta w_{14} = 0,003$,

$\Delta w_{24} = -0,003$

$W_{05}' = W_{05} + \Delta W_{05} = 2 + 0,032 = 2,032$
$W_{35}' = W_{35} + \Delta W_{35} = -1 + 0,023 = -0,977$
$W_{45}' = W_{45} + \Delta W_{45} = 1 + 0,004 = 1,004$
$W_{03}' = W_{03} + \Delta W_{03} = 1 - 0,006 = 0,994$
$W_{13}' = W_{13} + \Delta W_{13} = -1 + 0,006 = -0,994$
$W_{23}' = W_{23} + \Delta W_{23} = -1 - 0,006 = -1,006$
$W_{04}' = W_{04} + \Delta W_{04} = -2 + 0,003 = -1,997$
$W_{14}' = W_{14} + \Delta W_{14} = -1 - 0,003 = -1,003$
$W_{24}' = W_{24} + \Delta W_{24} = -1 + 0,003 = -0,997$

$\beta_3' = W_{03}' + x_1 W_{13}' + x_2 W_{23}' = 0,994 + 0,994 - 1,006 =$
$= 0,982$
$a_3' = g(\beta_3') = 0,728$
$\beta_4' = W_{04}' + x_1 W_{14}' + x_2 W_{24}' = -1,997 + 1,003 - 0,997 =$
$= -1,991$
$a_4' = g(\beta_4') = 0,12$
$\beta_5' = W_{05}' + a_3' W_{35}' + a_4' W_{45}' = 2,032 + 0,728(-0,977)$
$+ 0,12(1,004) = 1,441$
$a_5' = g(\beta_5') = 0,809$

$E' = \frac{1}{2}(t_5 - a_5')^2 = 0,018 < 0,02$

γ) $g(x) = \tanh(x)$

$$g'(x) = \frac{(e^x + e^{-x})^2}{(e^x + e^{-x})^2} - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2} = 1 - g^2(x)$$

$\beta_3 = W_{03} + x_1 W_{13} + x_2 W_{23} = 1$
$a_3 = g(\beta_3) = 0,762$
$\beta_4 = W_{04} + x_1 W_{14} + x_2 W_{24} = -2$
$a_4 = g(\beta_4) = -0,964$
$\beta_5 = W_{05} + a_3 W_{35} + a_4 W_{45} = 2 - 0,762 - 0,964 =$
$= 0,274$

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$a_5 = g(b_5) = 0,267$

$E = \frac{1}{2}(t_5 - a_5)^2 = 0,269$

$\Delta w_{05} = -\epsilon \frac{\partial E}{\partial b_5} = -(t_5 - a_5)\, g'(b_5) = -0,733 \cdot 0,929 =$ (g tanh)

$= -0,681$

$\Delta w_{35} = -\epsilon a_3 \frac{\partial E}{\partial b_5} = -0,762 \cdot 0,681 = -0,519$

$\Delta w_{45} = -\epsilon a_4 \frac{\partial E}{\partial b_5} = -0,964 \cdot 0,681 = 0,656$

$\Delta w_{03} = -\epsilon \frac{\partial E}{\partial b_3} = -0,681 \cdot w_{35}\, g'(b_3) = 0,286$

$\Delta w_{04} = -\epsilon \frac{\partial E}{\partial b_4} = -0,681 \cdot w_{45}\, g'(b_4) = -0,048$

$\Delta w_{13} = -0,286$, $\Delta w_{23} = 0,286$, $\Delta w_{14} = 0,048$,
$\Delta w_{24} = -0,048$

$w_{05}' = w_{05} - \Delta w_{05} = 2 + 0,681 = 2,681$
$w_{35}' = w_{35} - \Delta w_{35} = -1 + 0,519 = 0,481$
$w_{45}' = w_{45} - \Delta w_{45} = 1 - 0,656 = 0,344$
$w_{03}' = w_{03} - \Delta w_{03} = 1 - 0,286 = 0,714$
$w_{13}' = w_{13} - \Delta w_{13} = -1 + 0,286 = -0,714$
$w_{23}' = w_{23} - \Delta w_{23} = -1 - 0,286 = -1,286$
$w_{04}' = w_{04} - \Delta w_{04} = -2 + 0,048 = -1,952$
$w_{14}' = w_{14} - \Delta w_{14} = -1 - 0,048 = -1,048$
$w_{24}' = w_{24} - \Delta w_{24} = -1 + 0,048 = -0,952$

$b_3' = w_{03}' + x_1 w_{13}' + x_2 w_{23}' = 0,714 + 0,714 - 1,286 =$
$= 0,142$

$a_3' = g(b_3') = 0,141$

$b_4' = w_{04}' + x_1 w_{14}' + x_2 w_{24}' = -1,952 + 1,048 - 0,952 =$

$= -1,856$

$a_4' = g(b_4') = -0,952$

$b_5' = w_{05}' + a_3' w_{35}' + a_4' w_{45}' = 2,681 + 0,141 \cdot 0,481 - 0,959 \cdot 0,344 = 2,421$

$\boxed{a_5' = g(b_5') = 0,984}$

$\boxed{E' = \frac{1}{2}(t_5 - a_5')^2 = 0,0001} \ll 0,269$

Άσκηση 2 (Convolutional Neural Networks)

Διάσταση εικόνων: $369 \times 369 \times 3$ (RGB)

$W_1 = 369, \ H_1 = 369, \ D_1 = 3$

Διάσταση φίλτρου: $11 \times 11 \times 3$

$F = 11, \ K = 96, \ S = 3, \ P = 1$

α) $W_2 = \frac{(W_1 - F + 2 \cdot P)}{S} + 1 = \frac{369 - 11 + 2}{3} + 1 = 121$

$H_2 = \frac{(H_1 - F + 2P)}{S} + 1 = \frac{369 - 11 + 2}{3} + 1 = 121$

$D_2 = K = 96$

Επομένως οι διαστάσεις της εξόδου του 1ου convolutional layer είναι $\boxed{W_2 \times H_2 \times D_2 = 121 \times 121 \times 96}$

β) Ο αριθμός των units στο 1ο convolutional layer είναι $\boxed{121 \cdot 121 \cdot 96 = 1405536}$

γ) Ο αριθμός των εκπαιδεύσιμων παραμέτρων του 1ου convolutional layer με διαμοιρασμό βαρών, δηλαδή το μάθε ένα από τα 96 φίλτρα να έχει το δικό του βάρος, είναι $\boxed{96 \cdot (11 \cdot 11 \cdot 3 + 1) = 34944}$

δ) Αν αντικαθιστούσαμε το CNN με ένα Feed Forward layer με 256 νευρώνες, ο κάθε ένας από αυτούς θα λάμβανε ως είσοδο ολόκληρη την εικόνα, επομένως θα είχαμε $256 \cdot (369 \cdot 369 \cdot 3 + 1) =$
$= \boxed{104.571.904 \text{ εκπαιδεύσιμες παράμετροι}}$

Άσκηση 3 (Recurrent Neural Networks)

$C = \{0: \text{θετικό}, 1: \text{ουδέτερο}, 2: \text{αρνητικό}\}$

$h_0 = 0$

$W_{xh} = \begin{bmatrix} 0 & -1 & 2 \\ 1 & -2 & 0 \end{bmatrix}$, $W_{hy} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -1 \end{bmatrix}$

$W_{hh} = I_{2 \times 2}$

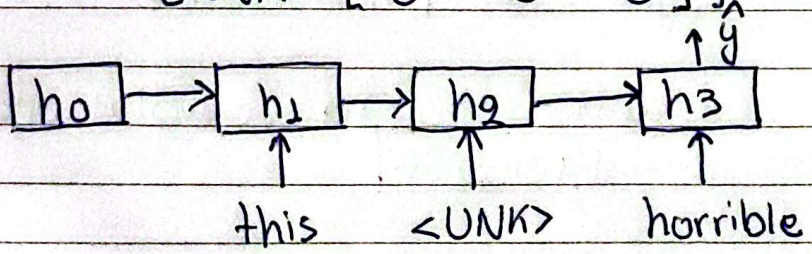
$b = 0$, $c = 0$

$g(x) = \max\{0, x\}$

$\hat{y} = h_3$

$V =$
this: [0 −1 2]
is: [1 −2 0]
good: [3 1 −2]
bad: [−2 −1 1]
great: [4 1 −2]
horrible: [−2 −2 1]
<UNK>: [0 0 0]

1.



$h_1 = g(W_{xh}x_1 + W_{hh}h_0 + c) = g\left(\begin{bmatrix} 0 & -1 & 2 \\ 1 & -2 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + \right.$

$\left. + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = g\left(\begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$

$$h_2 = g(W_{xh}x_2 + W_{hh}x_1 + c) = g\left(\begin{bmatrix} 0 & -1 & 2 \\ 1 & -2 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \right.$$

$$\left. + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = g\left(\begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$h_3 = g(W_{xh}x_3 + W_{hh}x_2 + c) = g\left(\begin{bmatrix} 0 & -1 & 2 \\ 1 & -2 & 0 \end{bmatrix}\begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} + \right.$$

$$\left. + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = g\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 2 \end{bmatrix}\right) = g\left(\begin{bmatrix} 9 \\ 4 \end{bmatrix}\right) =$$

$$= \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

$$\hat{y} = W_{hy}h_3 + b = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 9 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 14 \end{bmatrix}$$

Όταν περάσουμε την έξοδο του μοντέλου από μια Softmax, μεγαλύτερη πιθανότητα θα δοθεί στην τρίτη γραμμή, δηλαδή στην κλάση 2. Επομένως, το μοντέλο ταξινομεί την πρόταση "This was horrible" ως **αρνητική**

2. Το μοντέλο είναι καλά εκπαιδευμένο για το δοθέν δείγμα, καθώς οι τιμές των βαρών είναι κατάλληλες ώστε να ταξινομούν σωστά δείγματα με θετικά ή αρνητικά συναισθήματα. Αυτό ισχύει διότι εκμεταλλεύεται τα χαρακτηριστικά των δειγμάτων ανάλογα με το συναίσθημά τους, δηλαδή θετική 1η στήλη και αρνητική 2η στήλη για τις θετικές, ενώ αρνητική 1η στήλη και θετική 2η στήλη για τις αρνητικές. Ωστόσο, το μοντέλο δε θα απέδιδε καλά σε δείγματα που περιέχουν τη λέξη not, η οποία

λανθασμένα θα αντιμετωπίζονταν σαν <UNK> και δε θα επηρέαζε την ταξινόμηση της φράσης.

3. Υποθέτοντας ότι το μοντέλο αντί για το τελευταίο hidden state χρησιμοποιεί average pooling από τα h_1, h_2 και h_3, θα έχουμε:

$$\hat{y} = W_{hy}\frac{(h_1+h_2+h_3)}{3} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -1 \end{bmatrix} \frac{1}{3}\left(\begin{bmatrix} 5 \\ 2 \end{bmatrix} + \begin{bmatrix} 5 \\ 2 \end{bmatrix} + \begin{bmatrix} 9 \\ 4 \end{bmatrix}\right) =$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 2 & -1 \end{bmatrix} \frac{1}{3}\begin{bmatrix} 19 \\ 8 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 30 \end{bmatrix}\frac{1}{3} = \begin{bmatrix} 2,667 \\ 6,333 \\ 10 \end{bmatrix}$$

Επομένως, δε θα αλλάξει η πρόβλεψη για το ερώτημα 1.

Άσκηση 4 (Attention - Transformers)

$V = \{A, B, C, D, \text{<start>}, \text{<stop>}\}$

Παρατίθενται διορθώσεις των αποτελεσμάτων στο τέλος

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} A \\ B \\ C \\ D \\ \text{<start>} \\ \text{<stop>} \end{matrix}$$

$$W_Q = W_K = W_V = \begin{bmatrix} 0 & 3 \\ 1 & 2 \\ 2 & 1 \\ 3 & 0 \end{bmatrix}$$

a) $x_1 = D$, $x_2 = A$, $x_3 = A$, $x_4 = \langle start \rangle$, $x_5 = A$, $x_6 = D$, $x_7 = \langle stop \rangle$

$$Q_1 = E_{x_1} W_a = [0\ 0\ 0\ 1] \begin{bmatrix} 0 & 3 \\ 1 & 2 \\ 2 & 1 \\ 3 & 0 \end{bmatrix} = [3\ 0] = K_1 = V_1$$

$$Q_2 = E_{x_2} W_a = [1\ 0\ 0\ 0] \begin{bmatrix} 0 & 3 \\ 1 & 2 \\ 2 & 1 \\ 3 & 0 \end{bmatrix} = [0\ 3] = K_2 = V_2$$

$Q_3 = K_3 = V_3 = Q_2$.

$$Q_4 = E_{x_4} W_a = [0\ 0\ 0\ 0] \begin{bmatrix} 0 & 3 \\ 1 & 2 \\ 2 & 1 \\ 3 & 0 \end{bmatrix} = [0\ 0] = K_4 = V_4$$

$Q_5 = K_5 = V_5 = Q_2$

$Q_6 = K_6 = V_6 = Q_1$

$Q_7 = K_7 = V_7 = Q_4$

$$QK^T = S_{ij} = \begin{bmatrix} 9 & 0 & 0 & 0 & 0 & 9 & 0 \\ 0 & 9 & 9 & 0 & 9 & 0 & 0 \\ 0 & 9 & 9 & 0 & 9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 9 & 9 & 0 & 9 & 0 & 0 \\ 9 & 0 & 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

CamScanner

$S_{11} = Q_1 K_1^T = [3\ \ 0]\begin{bmatrix}3\\0\end{bmatrix} = 9 = S_{16} = S_{61} = S_{66}$

$S_{12} = Q_1 K_2^T = [3\ \ 0]\begin{bmatrix}0\\3\end{bmatrix} = 0 = S_{21} = S_{13} = S_{31} = S_{15} = S_{51}$

$= S_{26} = S_{62} = S_{36} = S_{63} = S_{56} = S_{65}$

$S_{14} = Q_1 K_4^T = [3\ \ 0]\begin{bmatrix}0\\0\end{bmatrix} = 0 = S_{41} = S_{24} = S_{42} = S_{34} =$

$= S_{43} = S_{45} = S_{54} = S_{46} = S_{64} = S_{17} = S_{71} = S_{27} = S_{72} = S_{37} =$

$= S_{73} = S_{57} = S_{75} = S_{67} = S_{76}$

$S_{23} = Q_2 K_3^T = [0\ \ 3]\begin{bmatrix}0\\3\end{bmatrix} = 9 = S_{32} = S_{26} = S_{62} = S_{36}$

$= S_{63} = S_{22} = S_{33} = S_{66}$

$S_{47} = Q_4 K_7^T = [0\ \ 0]\begin{bmatrix}0\\0\end{bmatrix} = 0 = S_{74} = S_{44} = S_{77}$

Ως d_K θα θεωρήσουμε τη μεγαλύτερη διάσταση του διανύσματος K, δηλαδή $d_K = 2$. Επομένως, $\sqrt{d_K} = \sqrt{2}$ και θα διαιρέσουμε τον πίνακα S_{ij} με $\sqrt{2}$.

Ο πίνακας προσοχής με χρήση scaled dot product attention προκύπτει ως

$A = \text{softmax}\left(\frac{S}{\sqrt{d_K}}\right)$, όπου $A_{ij} = \frac{\exp(S_{ij}/\sqrt{d_K})}{\sum_{k=1}^{7} \exp(S_{ik}/\sqrt{d_K})}$

$Sum_1 = e^{6,364} + 1 + 1 + 1 + 1 + e^{6,364} + 1 = 1166 = Sum_6$

$A_{11} = 580,5/1166 = 0,498 = A_{16} = A_{61} = A_{66}$

$A_{12} = 1/1166 = 0,001 = A_{13} = A_{14} = A_{15} = A_{17} = A_{62} = A_{63} = A_{64} =$

$= A_{65} = A_{67}$

$Sum_2 = 1 + 580,5 + 580,5 + 1 + 580,5 + 1 + 1 = 1743,5 =$
$= Sum_3 = Sum_5$

$A_{21} = 1/1745,5 = 0,001 = A_{24} = A_{26} = A_{27} = A_{31} = A_{34} = A_{36} =$
$A_{37} = A_{51} = A_{54} = A_{56} = A_{57}$

$A_{22} = 580,5/1745,5 = 0,333 = A_{23} = A_{25} = A_{32} = A_{33} = A_{35} =$
$= A_{52} = A_{53} = A_{55}.$

$Sum_4 = 1 + 1 + 1 + 1 + 1 + 1 + 1 = 7 = Sum_7$

$A_{41} = 1/7 = 0,143 = A_{42} = A_{43} = A_{44} = A_{45} = A_{46} = A_{47} = A_{71} =$
$= A_{72} = A_{73} = A_{74} = A_{75} = A_{76} = A_{77}$

$$A^{(1)} = \begin{bmatrix} 0,498 & 0,001 & 0,001 & 0,001 & 0,001 & 0,498 & 0,001 \\ 0,001 & 0,333 & 0,333 & 0,333 & 0,333 & 0,001 & 0,001 \\ 0,001 & 0,333 & 0,333 & 0,333 & 0,333 & 0,001 & 0,001 \\ 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 \\ 0,001 & 0,333 & 0,333 & 0,333 & 0,333 & 0,001 & 0,001 \\ 0,498 & 0,001 & 0,001 & 0,001 & 0,001 & 0,498 & 0,001 \\ 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 \end{bmatrix}$$

b)

$$W_Q^{(2)} = W_K^{(2)} = W_V^{(2)} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad W_0 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$Q_1^{(2)} = Ex_1 W_Q^{(2)} = [0\ 0\ 0\ 1] \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = [1\ 1] =$$

$= K_1^{(2)} = V_1^{(2)}$.

$$Q_2^{(2)} = Ex_2 W_Q^{(2)} = [1\ 0\ 0\ 0] \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = [1\ 1] = K_2^{(2)} =$$

$= V_2^{(2)}$

CamScanner

$Q_3^{(2)} = K_3^{(2)} = V_3^{(2)} = Q_2^{(2)}$

$$Q_4^{(2)} = E_{x4} W_a^{(2)} = [0\ 0\ 0\ 0] \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = [0\ 0] = K_4^{(2)} = V_4^{(2)}$$

$Q_5^{(2)} = K_5^{(2)} = V_5^{(2)} = Q_2^{(2)}$

$Q_6^{(2)} = K_6^{(2)} = V_6^{(2)} = Q_1^{(2)}$

$Q_7^{(2)} = K_7^{(2)} = V_7^{(2)} = Q_4^{(2)}$

$$S_{ij}^{(2)} = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 2 & 0 & 2 & 0 & 0 \\ 0 & 2 & 2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$S_{11}^{(2)} = Q_1^{(2)} K_1^{T(2)} = [1\ 1] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2 = S_{16}^{(2)} = S_{61}^{(2)} = S_{66}^{(2)} =$$

$= S_{12}^{(2)} = S_{21}^{(2)} = S_{13}^{(2)} = S_{31}^{(2)} = S_{15}^{(2)} = S_{51}^{(2)} = S_{26}^{(2)} = S_{62}^{(2)} =$

$= S_{36}^{(2)} = S_{63}^{(2)} = S_{56}^{(2)} = S_{65}^{(2)} = S_{23}^{(2)} = S_{32}^{(2)} = S_{22}^{(2)} = S_{33}^{(2)} =$

$= S_{66}^{(2)}$

$$S_{14}^{(2)} = Q_1^{(2)} K_4^{(2)T} = [1\ 1] \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 = S_{41}^{(2)} = S_{24}^{(2)} =$$

$= S_{42}^{(2)} = S_{34}^{(2)} = S_{43}^{(2)} = S_{45}^{(2)} = S_{54}^{(2)} = S_{46}^{(2)} = S_{64}^{(2)} = S_{14}^{(2)} =$

$= S_{71}^{(2)} = S_{27}^{(2)} = S_{72}^{(2)} = S_{37}^{(2)} = S_{73}^{(2)} = S_{57}^{(2)} = S_{75}^{(2)} =$

$= S_{67}^{(2)} = S_{76}^{(2)}$

$S_{47}^{(2)} = Q_4^{(2)} K_7^{(2)T} = [0 \ 0] \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 = S_{74}^{(2)} = S_{44}^{(2)} = S_{77}^{(2)}$

Ως δκ θεωρούμε πάλι τη μεγαλύτερη διάσταση του διανύσματος κ, αρα διαιρούμε τον πινακα Sij με $\sqrt{2}$.

$Sum_1^{(2)} = e^{1,414} + 1 + 1 + 1 + 1 + e^{1,414} + 1 = 13,225 = Sum_6^{(2)}$

$A_{11}^{(2)} = 4,112 / 13,225 = 0,311 = A_{16}^{(2)} = A_{61}^{(2)} = A_{66}^{(2)}$

$A_{12}^{(2)} = 1 / 13,225 = 0,076 = A_{13}^{(2)} = A_{14}^{(2)} = A_{15}^{(2)} = A_{17}^{(2)} =$

$= A_{62}^{(2)} = A_{63}^{(2)} = A_{64}^{(2)} = A_{65}^{(2)} = A_{67}^{(2)}$

$Sum_2^{(2)} = 1 + 4,112 + 4,112 + 1 + 4,112 + 1 + 1 = 16,336 =$

$= Sum_3^{(2)} = Sum_5^{(2)}$

$A_{21}^{(2)} = 1 / 16,336 = 0,061 = A_{24}^{(2)} = A_{26}^{(2)} = A_{27}^{(2)} = A_{31}^{(2)} =$

$= A_{34}^{(2)} = A_{36}^{(2)} = A_{37}^{(2)} = A_{51}^{(2)} = A_{54}^{(2)} = A_{56}^{(2)} = A_{57}^{(2)}$

$A_{22}^{(2)} = 4,112 / 16,336 = 0,252 = A_{23}^{(2)} = A_{25}^{(2)} = A_{32}^{(2)} =$

$= A_{33}^{(2)} = A_{35}^{(2)} = A_{52}^{(2)} = A_{53}^{(2)} = A_{55}^{(2)}$

$Sum_4^{(2)} = 1 + 1 + 1 + 1 + 1 + 1 + 1 = 7 = Sum_7^{(2)}$

$A_{41}^{(2)} = 1/7 = 0,143 = A_{42}^{(2)} = A_{43}^{(2)} = A_{44}^{(2)} = A_{45}^{(2)} = A_{46}^{(2)} =$

$= A_{47}^{(2)} = A_{71}^{(2)} = A_{72}^{(2)} = A_{73}^{(2)} = A_{74}^{(2)} = A_{75}^{(2)} = A_{76}^{(2)} = A_{77}^{(2)}$

Επομένως ο πινακας προσοχής με χρήση scaled dot product attention για τη δεύτερη κεφαλή προσοχής είναι:

$$A^{(2)} = \begin{bmatrix} 0,311 & 0,076 & 0,076 & 0,076 & 0,076 & 0,311 & 0,076 \\ 0,061 & 0,252 & 0,252 & 0,061 & 0,252 & 0,061 & 0,061 \\ 0,061 & 0,252 & 0,252 & 0,061 & 0,252 & 0,061 & 0,061 \\ 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 \\ 0,061 & 0,252 & 0,252 & 0,061 & 0,252 & 0,061 & 0,061 \\ 0,311 & 0,076 & 0,076 & 0,076 & 0,076 & 0,311 & 0,076 \\ 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 & 0,143 \end{bmatrix}$$

$$Z_1 = A^{(1)} V^{(1)} = A^{(1)} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 3 \\ 0 & 3 \\ 0 & 0 \\ 0 & 3 \\ 3 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2,988 & 0,009 \\ 0,006 & 2,997 \\ 0,006 & 2,997 \\ 0,858 & 1,287 \\ 0,006 & 2,997 \\ 2,988 & 0,009 \\ 0,858 & 1,287 \end{bmatrix}$$

$$Z_2 = A^{(2)} V^{(2)} = A^{(2)} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0,85 & 0,85 \\ 0,878 & 0,878 \\ 0,878 & 0,878 \\ 0,715 & 0,715 \\ 0,878 & 0,878 \\ 0,85 & 0,85 \\ 0,715 & 0,715 \end{bmatrix}$$

$$\text{Multihead}(x) = \text{Concat}(Z_1, Z_2) \cdot W_0 =$$

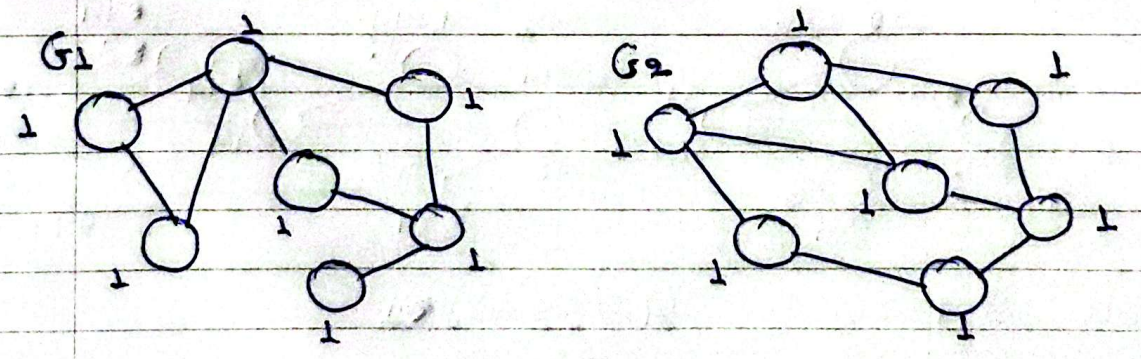
$$= \begin{bmatrix} 2,988 & 0,009 & 0,85 & 0,85 \\ 0,006 & 2,997 & 0,878 & 0,878 \\ 0,006 & 2,997 & 0,878 & 0,878 \\ 0,858 & 1,287 & 0,715 & 0,715 \\ 0,006 & 2,997 & 0,878 & 0,878 \\ 2,988 & 0,009 & 0,85 & 0,85 \\ 0,858 & 1,287 & 0,715 & 0,715 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

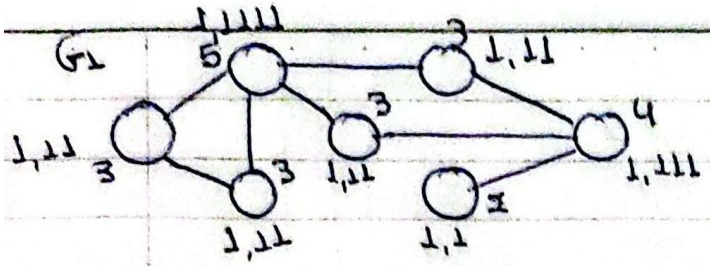
CamScanner

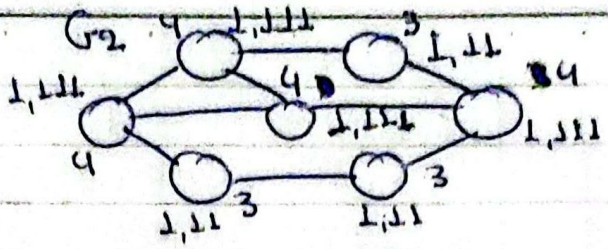
$$=\begin{bmatrix} 0,859 & 3,838 & 0,859 & 3,838 \\ 3,875 & 0,884 & 3,875 & 0,884 \\ 3,875 & 0,884 & 3,875 & 0,884 \\ 2,002 & 1,573 & 2,002 & 1,573 \\ 3,875 & 0,884 & 3,875 & 0,884 \\ 0,859 & 3,838 & 0,859 & 3,838 \\ 2,002 & 1,573 & 2,002 & 1,573 \end{bmatrix}$$

Το μοντέλο μας είναι decoder-only, επομένως δεν πρέπει να μπορεί να βλέπει στοιχεία της ακολουθίας που παράγονται αργότερα, μόνο αυτά που έχουν παραχθεί ήδη. Επομένως, πριν από την εφαρμογή της softmax, στους πίνακες $S_{ij}^{(1)}$ και $S_{ij}^{(2)}$ θα πρέπει να εφαρμόσουμε masking, δηλαδή τα στοιχεία πάνω από την κύρια διαγώνιο να τα θέσουμε ίσα με $-\infty$. Τα στοιχεία αυτά στους πίνακες $A^{(1)}$ και $A^{(2)}$ θα είναι μηδέν και τα υπόλοιπα θα τροποποιηθούν κατάλληλα. Η ιδέα που παρουσιάσα είναι για encoder models, αλλά εφαρμόζοντας τα παραπάνω μετά από κατάλληλες διορθώσεις προκύπτει το σωστό αποτέλεσμα.

Άσκηση 5 (Graph Neural Networks)





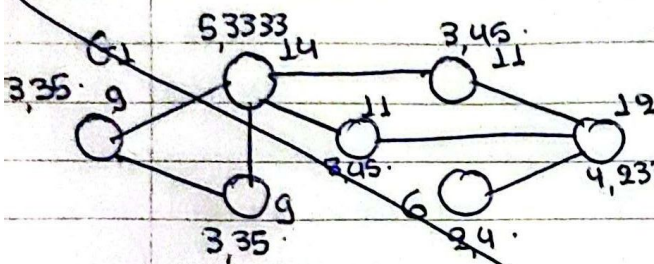


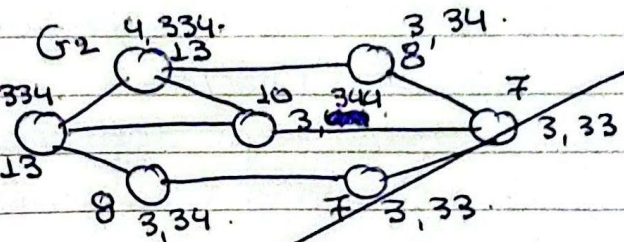
HASH $1,1 \rightarrow 2$

$1,11 \rightarrow 3$

$1,111 \rightarrow 4$

$1,1111 \rightarrow 5$



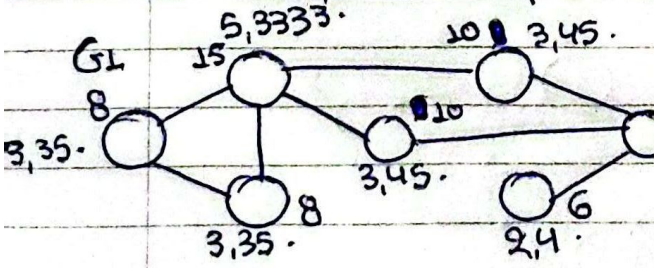


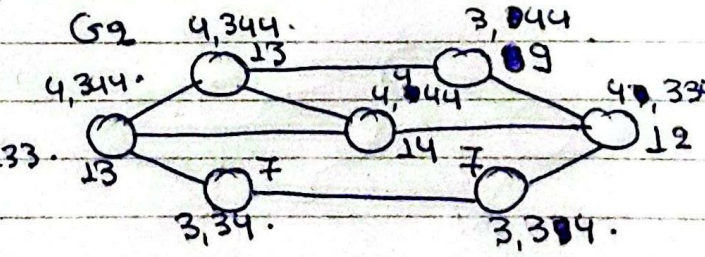
HASH $2,4 \rightarrow 6$ $3,34 \rightarrow 8$ $3,35 \rightarrow 9$ $3,344 \rightarrow 10$ $3,44 \rightarrow 11$ $3,45 \rightarrow 12$ $4,233 \rightarrow 13$ $4,334 \rightarrow 14$ $5,3333 \rightarrow 15$

$\varphi(G_1) = [7,1,4,1,1,1,0,0,2,0,2,1,0,1]$

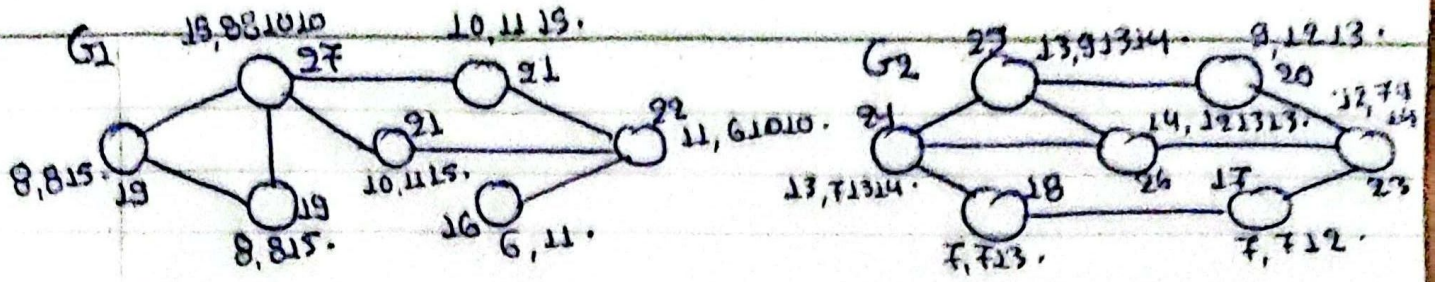
$\varphi(G_2) = [7,0,5,2,0,0,2,2,0,1,0,0,2,0]$

$K(G_1, G_2) = \varphi(G_1)^{T} \varphi(G_2) = 71$





HASH $2,4 \rightarrow 6$ $3,34 \rightarrow 7$ $3,35 \rightarrow 8$ $3,44 \rightarrow 9$ $3,45 \rightarrow 10$ $4,233 \rightarrow 11$ $4,3334 \rightarrow 12$ $4,344 \rightarrow 13$ $4,444 \rightarrow 14$ $5,3333 \rightarrow 15$



HASH

6,11 → 16
7,7,12 → 17
7,7,13 → 18
8,8,15 → 19
9,9,12,13 → 20
10,11,15 → 21
11,6,10,10 → 22
12,7,9,14 → 23
13,7,13,14 → 24
13,9,13,14 → 25
14,12,13,13 → 26
15,8,8,10 → 27

$\varphi(G_1) = [7,1,4,1,1,1,0,2,0,2,1,0,0,0,1,1,0,0,2,0,2,1,0,0,0,0,1]$

$\varphi(G_2) = [7,0,3,4,0,0,2,0,1,0,0,1,2,1,0,0,1,1,0,1,0,0,1,1,1,0]$

$\boxed{K(G_1,G_2) = (\varphi(G_1))^T \varphi(G_2) = 65}$